

Mitochondria are double membrane-bound cellular organelles that house several connected metabolic reactions. The citric acid cycle takes place inside the mitochondrial matrix and produces two reduced electron carriers, NADH and FADH₂. These two carriers then pass their electrons to ubiquinone (UQ), which is reduced to ubiquinol (UQH₂). UQH₂ passes its electrons to oxidized cytochrome C (cyt-C_{ox}), regenerating UQ and forming reduced cytochrome C (cyt-C_{red}). Finally, cyt-C_{red} transfers electrons to oxygen, regenerating cyt-C_{ox} and forming water. Each reaction is facilitated by one of four protein complexes that are collectively known as the electron transport chain (ETC).

Researchers isolated mitochondria and treated them with a novel ETC inhibitor, "drug X." After an overnight incubation, they measured levels of several ETC-reduced electron carriers in the presence of drug X relative to untreated mitochondria. Their results are shown in Figure 1.

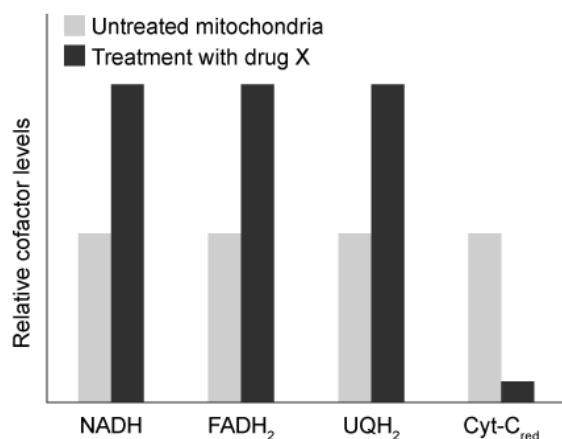
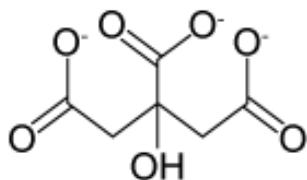


Figure 1 Relative levels of reduced electron carriers in the presence and absence of drug X

In a second experiment, the researchers exposed mitochondria to FCCP, a chemical that can passively transport protons across membranes. They observed that in the presence of FCCP, the citric acid cycle was fully active and all reduced cofactors were present at the same levels as in untreated mitochondria. However, the rate of mitochondrial ATP synthesis was significantly reduced. ATP synthesis was immediately restored when FCCP was washed away.

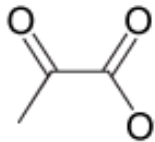
Which of the following metabolites is NOT produced in one of the metabolic pathways discussed in the passage?

☐ A.



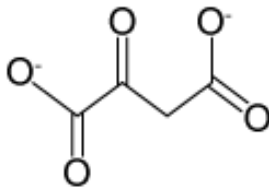
[39%]

✓ ☒ B.



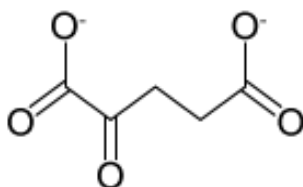
[39%]

☐ C.



[9%]

☐ D.

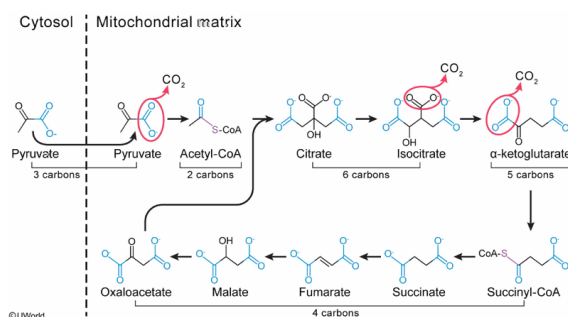


[11%]

Correct

39% answered correctly

Explanation:



Many catabolic pathways occur in the mitochondria. The passage discusses three of these pathways: the **citric acid cycle**, the **electron transport chain**, and **mitochondrial ATP synthesis** (ie, oxidative phosphorylation). The diagrams display several metabolites (ie, molecules produced or consumed in metabolic reactions), and the question asks which is NOT a part of these pathways.

With the exceptions of acetyl-CoA and succinyl-CoA, molecules in the citric acid cycle can be easily identified by the fact that they are all **flanked by carboxylate ions** on both ends of the main carbon chain. Succinyl-CoA and acetyl-CoA can be identified by their **thioesters** (S-CoA). Three of the given molecules meet these criteria and are part of the citric acid cycle:

- **Choice A** shows citrate, a six-carbon molecule that is synthesized in the first step of the citric acid cycle when acetyl-CoA combines with oxaloacetate.
- **Choice C** shows oxaloacetate, a four-carbon molecule that forms in the last step of the citric acid cycle when malate is oxidized.

- **Choice D** shows α -ketoglutarate, a five-carbon molecule that forms in the citric acid cycle when isocitrate releases CO_2 .

Choice B shows **pyruvate**, a three-carbon molecule produced *in the cytosol* after the final step of glycolysis. After pyruvate is synthesized, it is transported to the mitochondria, where it is decarboxylated to form acetyl-CoA. Acetyl-CoA then enters the citric acid cycle; pyruvate is not part of any of the pathways mentioned in the passage.

Educational objective:

Glycolysis occurs in the cytosol and ends with the synthesis of pyruvate. Pyruvate is transported into the mitochondria, where it is decarboxylated to form acetyl-CoA, which then enters the citric acid cycle. All steps of the citric acid cycle occur in the mitochondria and, unlike glycolysis, involve thioesters or multiple carboxylate groups.

Time spent:0 sec

Subject:
Biochemistry

QID:402231

System:
Metabolic Reactions

Mitochondria are double membrane-bound cellular organelles that house several connected metabolic reactions. The citric acid cycle takes place inside the mitochondrial matrix and produces two reduced electron carriers, NADH and FADH_2 . These two carriers then pass their electrons to ubiquinone (UQ), which is reduced to ubiquinol (UQH_2). UQH_2 passes its electrons to oxidized cytochrome C (cyt-C_{ox}), regenerating UQ and forming reduced cytochrome C ($\text{cyt-C}_{\text{red}}$). Finally, $\text{cyt-C}_{\text{red}}$ transfers electrons to oxygen, regenerating cyt-C_{ox} and forming water. Each reaction is facilitated by one of four protein complexes that are collectively known as the electron transport chain (ETC).

Researchers isolated mitochondria and treated them with a novel ETC inhibitor, "drug X." After an overnight incubation, they measured levels of several ETC-reduced electron carriers in the presence of drug X relative to untreated mitochondria. Their results are shown in Figure 1.

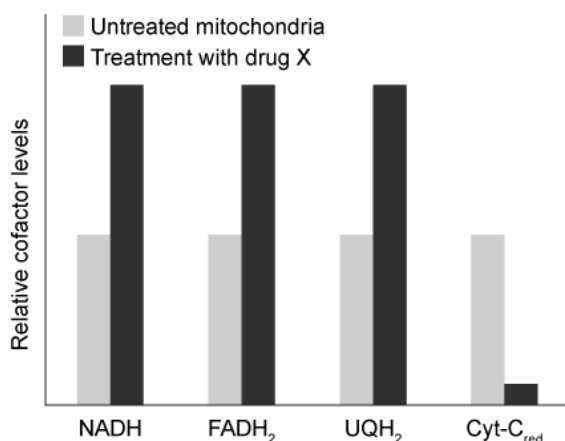


Figure 1 Relative levels of reduced electron carriers in the presence and absence of drug X

In a second experiment, the researchers exposed mitochondria to FCCP, a chemical that can passively transport protons across membranes. They observed that in the presence of FCCP, the citric acid cycle was fully active and all reduced cofactors were present at the same levels as in untreated mitochondria. However, the rate of mitochondrial ATP synthesis was significantly reduced. ATP synthesis was immediately restored when FCCP was washed away.

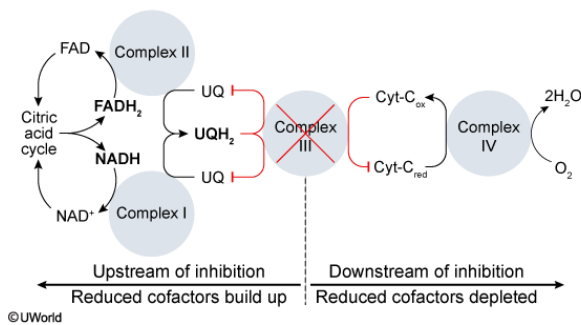
Based on Figure 1, which electron transport chain complex is inhibited by drug X?

- ☐ A. Complex I [2%]
- ☐ B. Complex II [6%]
- ☒ C. Complex III [52%]
- ☐ D. Complex IV [38%]

Correct

52% answered correctly

Explanation:



The electron carriers of the ETC normally exist in a **steady state**, produced as quickly as they are consumed. Inhibition of an ETC complex alters this steady state. Normally, complexes I and II pass electrons from NADH and FADH₂ to ubiquinone (UQ), producing ubiquinol (UQH₂). In **complex III**, UQH₂ then passes the electrons to oxidized cytochrome C (cyt-C_{ox}), **yielding reduced cytochrome C (cyt-C_{red})** and regenerating UQ. Finally, cyt-C_{red} passes its electrons to oxygen, yielding water and regenerating cyt-C_{ox}.

Figure 1 shows that drug X causes an accumulation of UQH₂ and its upstream electron carriers, NADH and FADH₂. Drug X also causes a **depletion of cyt-C_{red}**. If **complex III is inhibited**, it can no longer consume UQH₂ and can no longer produce cyt-C_{red}. The concentration of **UQH₂ builds** as complexes I and II continue to produce it while complex III fails to consume it. As UQH₂ levels increase, UQ levels fall, so eventually **NADH and FADH₂ will also accumulate** as they have nothing to pass their electrons to. Because inhibition of complex III means cyt-C_{ox} is no longer being reduced, **cyt-C_{red} is depleted** over time as complex IV continues to consume it. Therefore, drug X most likely inhibits complex III.

(Choice A) Inhibition of complex I would result in the buildup of NADH and the depletion of the downstream electron carriers.

(Choice B) Inhibition of complex II would result in the depletion of the downstream electron carriers UQH₂ and cyt-C_{red}, not cyt-C_{red} alone.

(Choice D) If complex IV were inhibited, all the reduced electron carriers would eventually build up, including cyt-C_{red}, because each would ultimately lose the ability to transfer its electrons to the next acceptor.

Educational objective:

Reduced electron carriers of the ETC are normally in a steady state, being consumed and regenerated as they move from one complex to another and back. Inhibiting a complex disturbs the steady state, causing an accumulation of upstream carriers and a depletion of downstream carriers.

Time spent:0 sec

Subject:
Biochemistry

QID:402232

System:
Metabolic Reactions

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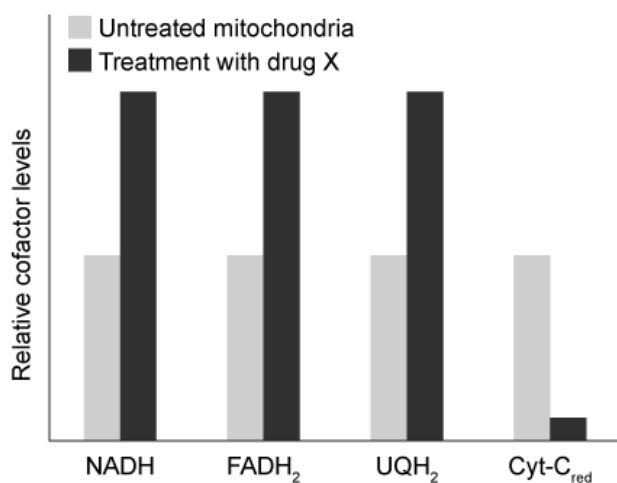


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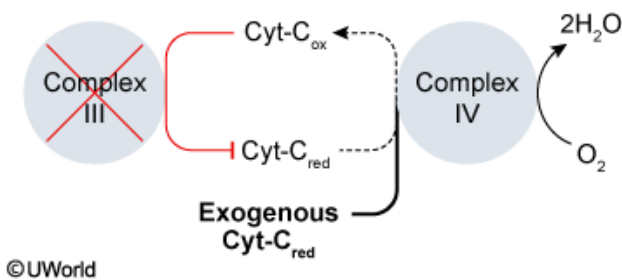
Which of the following could researchers add to restore oxygen consumption in the presence of drug X?

- ✔ ☒ A. cyt-C_{red} [61%]
- ☐ B. cyt-C_{ox} [24%]
- ☐ C. Ubiquinol [7%]
- ☐ D. Ubiquinone [6%]

Correct

62% answered correctly

Explanation:



Oxygen consumption in the electron transport chain takes place in **complex IV**, where O₂ gets reduced to become water. The electrons needed to reduce oxygen are provided by **reduced cytochrome C** (cyt-C_{red}), which is converted to its oxidized form (cyt-C_{ox}) in the reaction catalyzed by complex IV.

Figure 1 shows that cyt-C_{red} is depleted in the presence of drug X. Because cyt-C_{red} is required for complex IV to function, cyt-C_{red} depletion eventually results in decreased complex IV activity and therefore decreased oxygen consumption. To restore normal oxygen consumption levels, more cyt-C_{red} is needed. Therefore, researchers could add more cyt-C_{red} to restore normal oxygen consumption levels.

(Choice B) Oxidized cytochrome C (cyt-C_{ox}) is a *product* of oxygen consumption. It cannot be used to reduce oxygen to water, so adding more cyt-C_{ox} will not restore oxygen consumption levels.

(Choice C) Ubiquinol provides the electrons necessary for complex III to produce cyt-C_{red}, which in turn can reduce O₂. However, in the presence of drug X, complex III is inactive and cannot use electrons from ubiquinol to produce cyt-C_{red} no matter how much ubiquinol is added. Without cyt-C_{red}, oxygen consumption will not be restored.

(Choice D) Ubiquinone can be reduced to ubiquinol by complexes I and II. The generated ubiquinol would not be able to reduce cyt-C because complex III is inactive in the presence of drug X. Without cyt-C_{red}, oxygen consumption will not be restored.

Educational objective:

Each reduced electron carrier in the ETC facilitates reduction of the next acceptor in the chain. Depletion of one reduced electron carrier (ie, cyt-C_{red}), prevents reduction of the next in the chain (ie, oxygen). Adding more of a depleted carrier will reinstate activity until the carrier is depleted again.

Time spent:0 sec

Subject:
Biochemistry

QID:402233

System:
Metabolic Reactions

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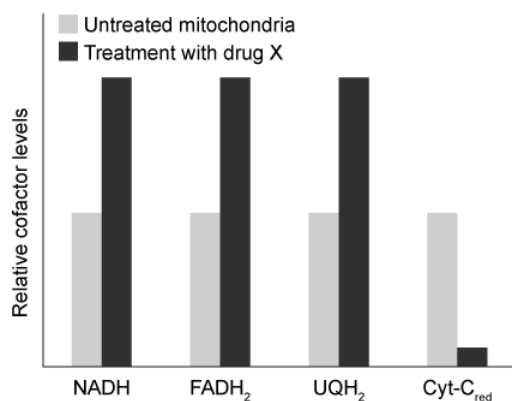


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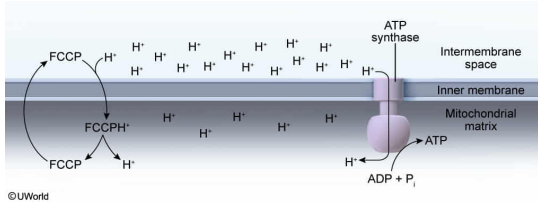
Based on the information in the passage, FCCP most likely causes a decrease in energy production by:

- ☒ A. decoupling the movement of protons down their concentration gradient from ATP synthase. [66%]
- ☐ B. carrying more protons into the intermembrane space than are pumped by the ETC alone. [9%]
- ☐ C. shifting the proton gradient further away from equilibrium than is achieved by the ETC alone. [8%]
- ☐ D. causing a decrease in ETC activity, resulting in fewer available protons for ATP synthase. [16%]

Correct

66% answered correctly

Explanation:



The electron transport chain (ETC) produces a pH difference across the inner mitochondrial membrane as protons are **pumped** from the matrix to the intermembrane space. These protons can then flow back into the matrix through ATP synthase. The movement of protons down their **electrochemical gradient through ATP synthase** provides the energy needed to produce ATP.

The passage states that FCCP can passively transport protons across membranes. Therefore, the addition of FCCP provides an alternative to ATP synthase by which protons can cross the membrane **down their concentration gradient** (ie, from high concentration to low concentration). Because the proton concentration is normally higher outside the mitochondrial matrix than inside it, FCCP transports protons into the matrix.

However, protons that enter the mitochondrial matrix by this means *do not interact* with ATP synthase, so they **do not drive ATP synthesis**; this phenomenon is called "**decoupling**." Because this decoupling results in fewer protons going through ATP synthase, the rate of **ATP synthesis decreases** in the presence of FCCP.

(Choice B) The movement of protons from the matrix to the intermembrane space would require FCCP to move protons against their concentration gradient (ie, active transport); however the passage states that FCCP-mediated transport is passive. In addition, more protons in the intermembrane space would *increase* the rate of ATP synthesis as more protons would be available to flow through ATP synthase.

(Choice C) FCCP brings the proton gradient closer to equilibrium because it only carries protons down their concentration gradient (ie, passive transport). Protons can only move against the gradient (ie, further from equilibrium) with the aid of energy input, as in the ETC.

(Choice D) The passage states that the citric acid cycle is fully active in the presence of FCCP, and that reduced electron carriers are neither depleted nor accumulated. Therefore, the ETC must be fully active and able to consume the reduced electron carriers from the citric acid cycle normally.

Educational objective:

ATP synthesis is driven by protons crossing the inner mitochondrial membrane through ATP synthase. When protons are transported across the membrane by means that bypass ATP synthase, the energy released in the process cannot be used to produce ATP. This phenomenon is known as "decoupling" because proton transfer is no longer coupled to ATP synthesis.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402234

System:
Metabolic Reactions